Rainul Hakim

rahakim@davidson.edu | Davidson, NC | linkedin.com/in/rainulhakim | github.com/RainulHakim | rainulhakim.vercel.app

Education

Davidson College — Davidson, NC
Bachelor of Science, Computer Science | GPA: 3.76 — *Expected May 2029*
- Relevant Coursework: Data Structures, Database Programming, Linear Algebra, Calculus I & Modeling, Calculus II

Technical Skills

Languages: Python, Java, SQL, R, JavaScript/TypeScript, C, HTML/CSS
ML/Data Science: PyTorch, PyTorch Geometric, TensorFlow.js, Scikit-learn, OpenCV, MediaPipe, NumPy, Pandas, Matplotlib
Web & Backend: React, Next.js, Node.js, REST APIs, Vercel Serverless Functions, Supabase, PostgreSQL
Tools: Git, GitHub, AWS, Linux, HPC Clusters, Mapbox, Quarto, VS Code

Experience

Machine Learning Research Assistant — May 2026 – Aug 2026
Facility for Rare Isotope Beams (FRIB) — *East Lansing, MI*
- Reached **88.82%** accuracy / **88.60%** weighted F1 on 3-class track classification by pretraining a **DGCNN** encoder with **SimCLR**-style contrastive learning on **60,795** unlabeled AT-TPC events, then fine-tuning via linear probing
- Beat augmentation baselines with a physically motivated rotation scheme, validated by ablation studies
- Trained with **PyTorch** and **PyTorch Geometric** on **HPC**; documented code for a cross-institutional codebase

Tech Consultant — Aug 2026 – Present
DataCATS (Consulting, Analytics, Tutoring Services), Davidson College — *Davidson, NC*
- Led **85**+ sessions and projects for students and faculty on data analysis, modeling, and experimental design
- Debug code and data pipelines and guide regression analysis, hypothesis testing, and visualization

Student Consultant — Feb 2026 – May 2026
The Jay Hurt Hub for Innovation and Entrepreneurship — *Davidson, NC*
- Built **Next.js/React** front end and **TypeScript** API routes for an AI study planner that drew faculty interest
- Cut malformed AI output with **Zod**-validated schemas and wellbeing guardrails, verified via iterative QA

Projects

Campus Event Discovery Platform | ***Python, PostgreSQL, AWS Lambda, Amazon RDS, API Gateway, Mapbox***
- Built event discovery platform ingesting Davidson event data into a relational **PostgreSQL** database
- Automated AWS pipeline with **EventBridge** and **Lambda** to extract, clean, and update event and location data
- Developed **React/Mapbox** interface to filter events by location, date, and category with campus navigation

Left No Crumbs! – 1st Place ($1,000), IdeaSprint | ***Next.js, Supabase, PostgreSQL, Web Push***
- Co-founded and built a real-time food-rescue platform alerting **600**+ students across **55** organizations to surplus catered meals
- Designed a **PostgreSQL** schema with row-level security, campus-only auth, dietary filtering, and rate-limited push notifications; partnered with Davidson Dining and T&I on Davidson One integration

AI Meal Analysis & Recommendation Platform | ***Python, Gemini API, React, Vite, JavaScript, Vercel***
- Built AI meal analysis pipeline generating macro breakdowns, health flags, and personalized alternatives
- Integrated **Gemini API** behind serverless functions and implemented meal comparison and weekly scoring systems
- Deployed React application used by **100**+ Davidson students to evaluate and compare campus meals

In-Browser Hand Gesture Recognition (ML) | ***JavaScript, TensorFlow.js, MediaPipe***
- Built a gesture-triggered effects app running fully in-browser at ∼**30 fps** with no backend: **MediaPipe Holistic** extracts **126** hand-landmark coordinates per frame and feeds a **TensorFlow.js** neural network
- Normalized landmarks by wrist position and scale for hand-size invariance; trained a binary classifier (Dropout **0.3**, **50** epochs, **0.97** confidence threshold) with a custom in-browser training UI
- Segmented the user's silhouette with **MediaPipe Selfie Segmentation** and rendered clones with smoke effects on **Canvas**

Leadership & Involvement

Resident Advisor, Davidson College — 2026 – Present
Logistics, Hack@Davidson — 2026 – Present
Event Coordinator, Davidson Entrepreneurship Club — 2025 – Present
Secretary, Association for Computing Machinery (ACM), Davidson College — 2026 – Present
Treasurer, Davidson International Association — 2026 – Present